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Parida

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(54) **COLD PLATE WITH COMBINED INCLINED IMPINGEMENT AND RIBBED CHANNELS**

2924/00 (2013.01); H01L 2924/0002 (2013.01); Y10T 29/4935 (2015.01); Y10T 29/49359 (2015.01)

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USPC 29/890.03, 890.035; 361/699, 702, 703; 165/80.3, 80.4
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B23P 15/26 (2006.01)

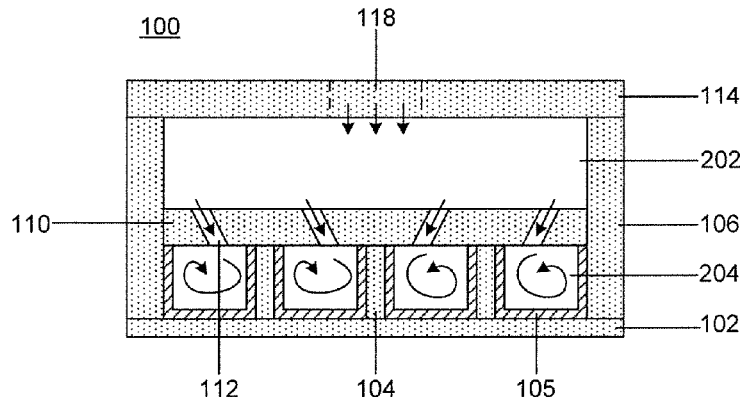
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(57) **ABSTRACT**

Methods for making heat transfer devices include forming a jet plate with a plurality of inclined jets set at an angular deviation from normal. A bottom plate is formed with channel walls that have ribs. The jet plate is attached to the bottom plate to form ribbed channels. The angular deviation of each inclined jet establishes a jet direction perpendicular to a long dimension of the ribbed channels.

8 Claims, 8 Drawing Sheets



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F28F 3/04 (2006.01)
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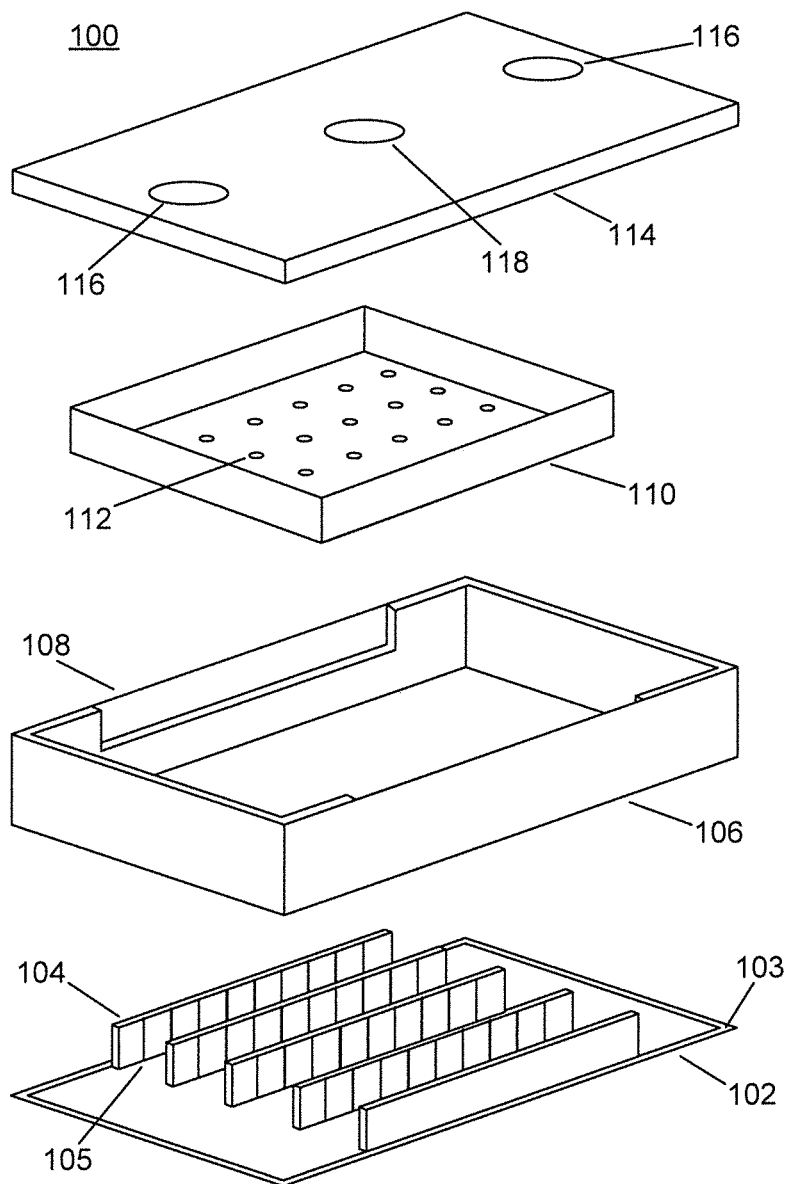


FIG. 1

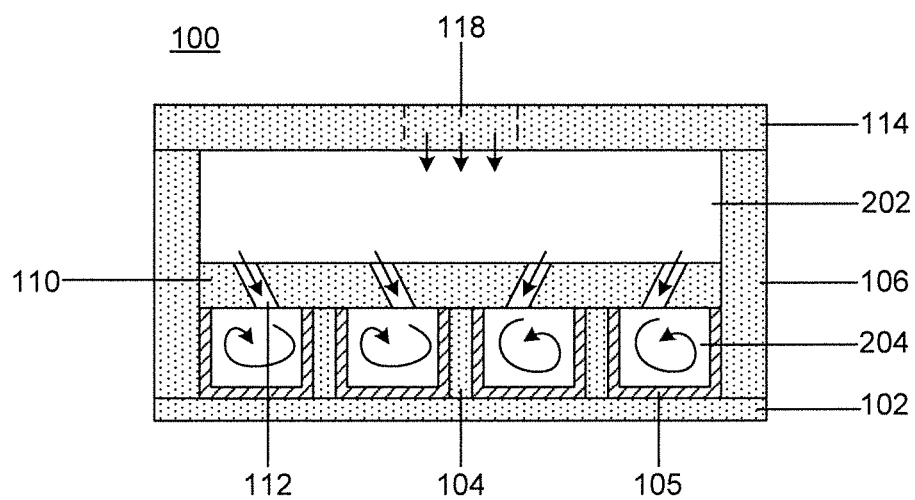


FIG. 2

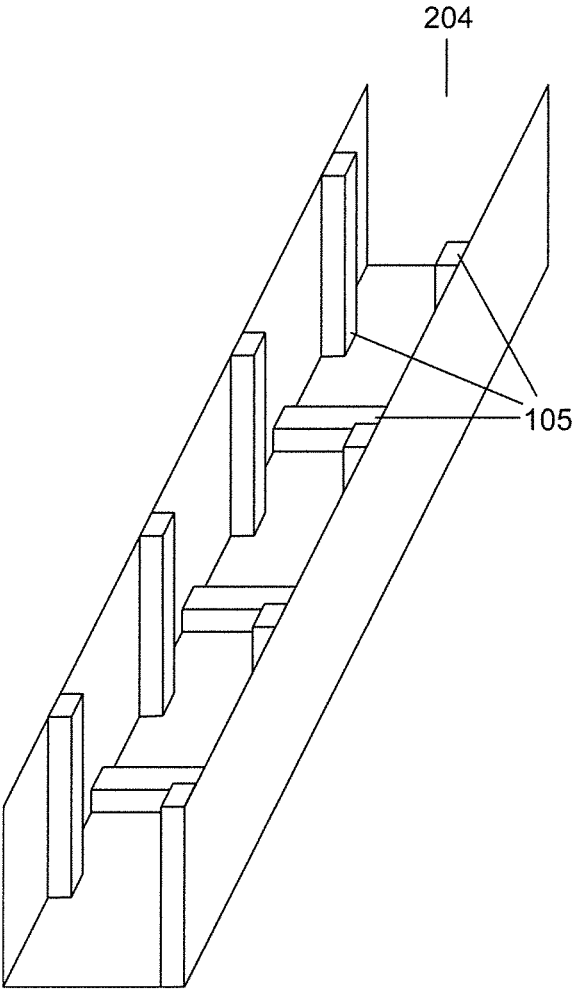


FIG. 3

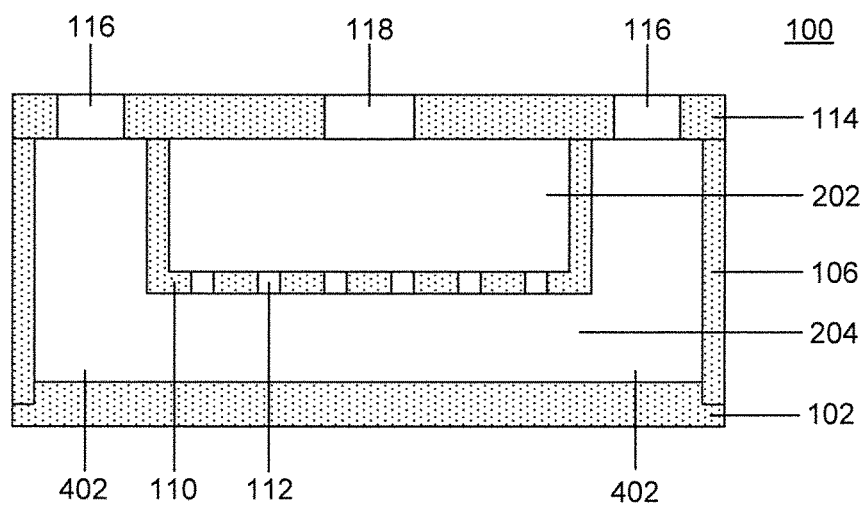


FIG. 4

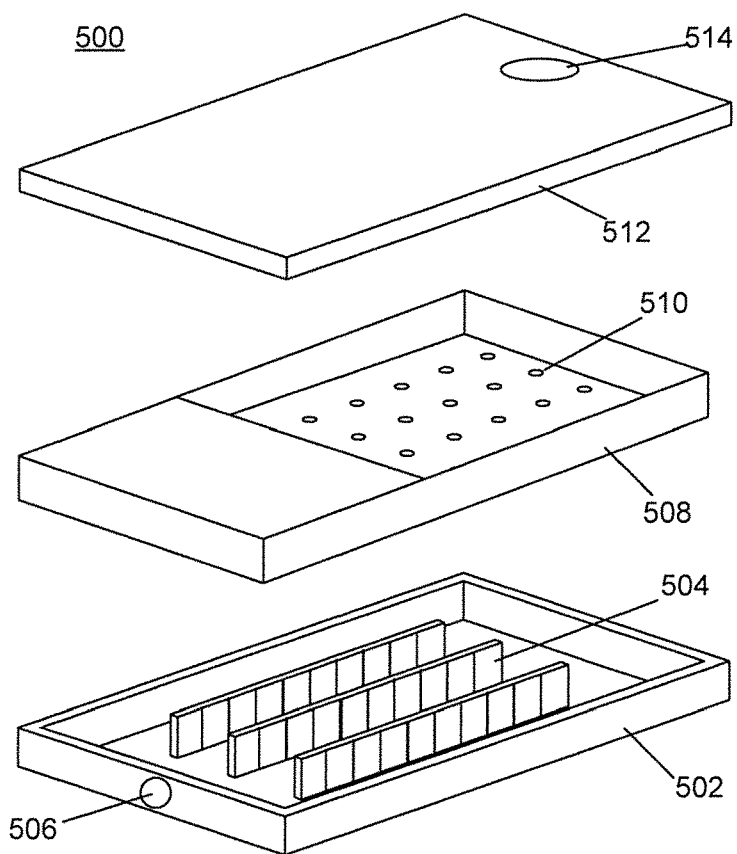


FIG. 5

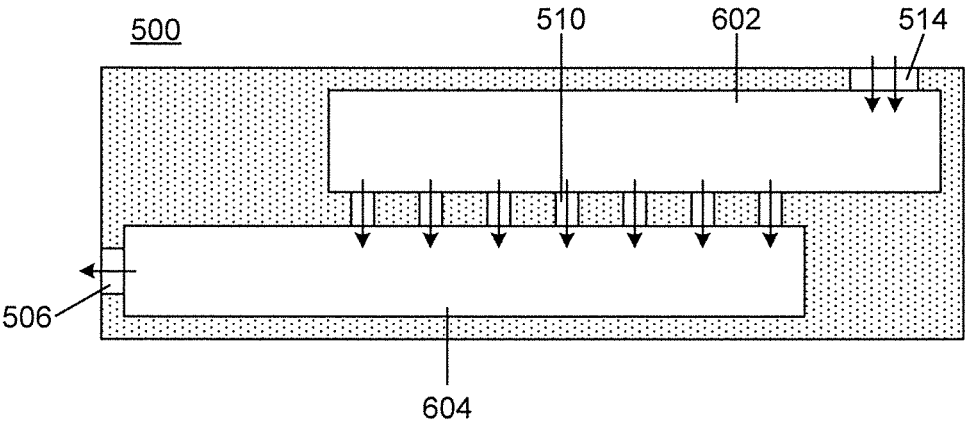


FIG. 6

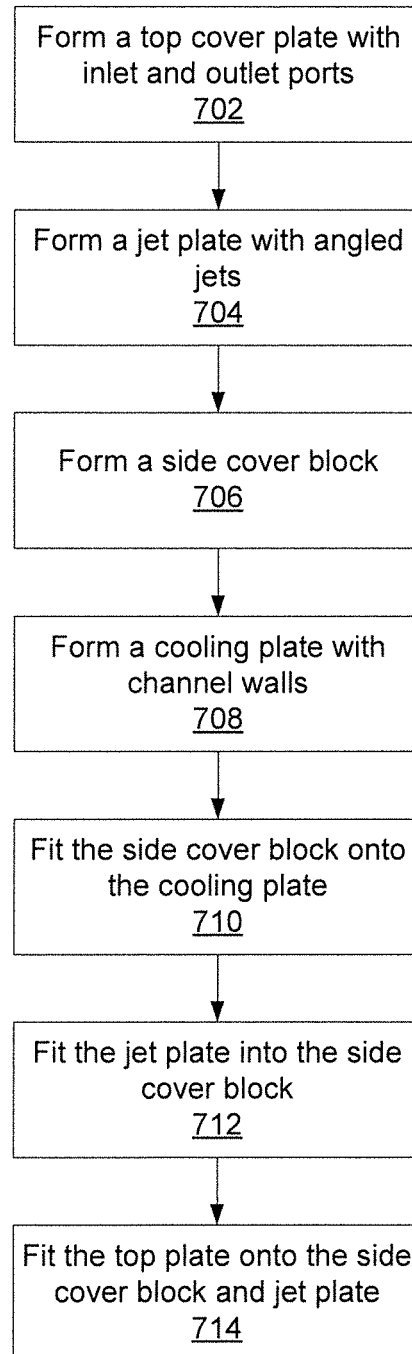


FIG. 7

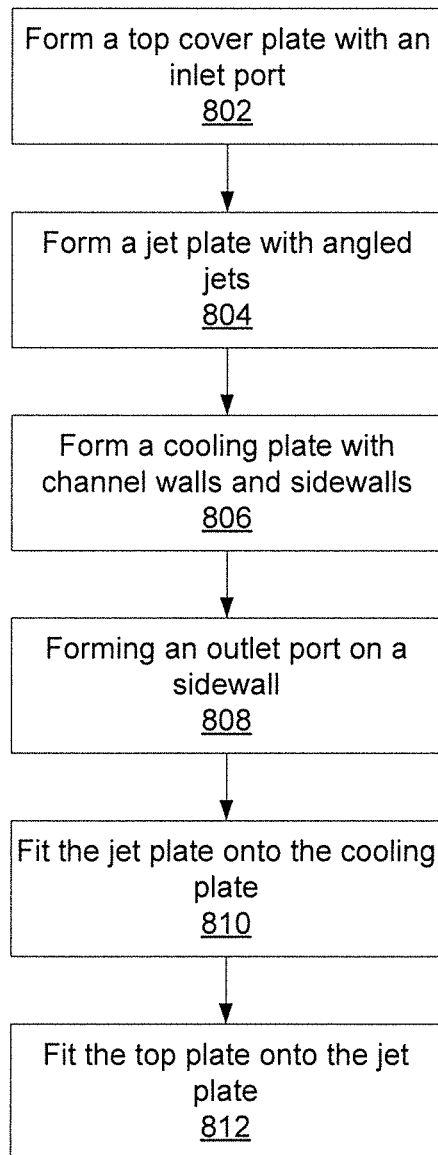


FIG. 8

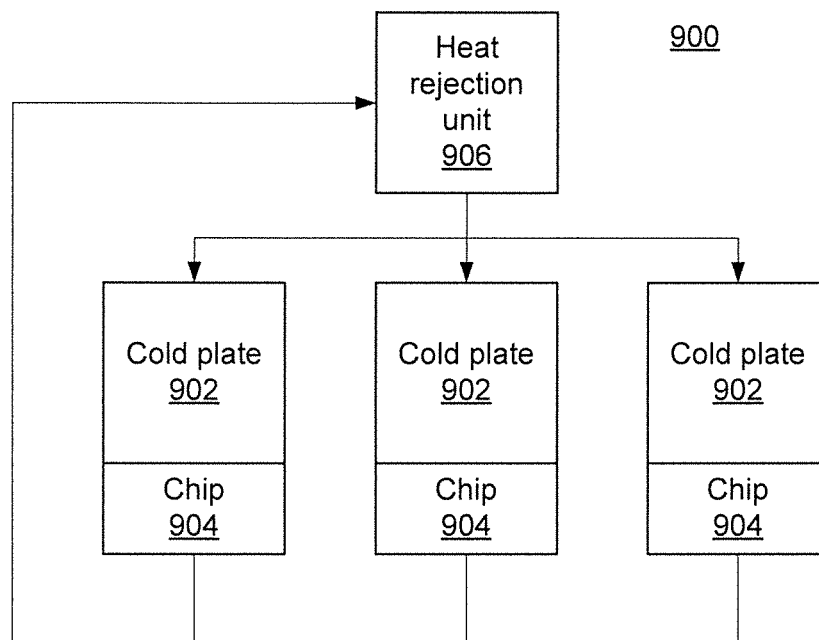


FIG. 9

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**COLD PLATE WITH COMBINED INCLINED
IMPINGEMENT AND RIBBED CHANNELS****GOVERNMENT RIGHTS**

This invention was made with Government support under Contract No.: DE-EE0002894 (Department of Energy). The government has certain rights in this invention.

BACKGROUND**Technical Field**

The present invention relates to cooling and, more particularly, to cooling devices that use inclined impingement of coolant fluid into ribbed channels.

Description of the Related Art

The demand for increased power and performance in semiconductor devices grows constantly. Meeting that demand produces an increased amount of heat, and therefore also increases the need for effective heat dissipation. While conventional cooling solutions have worked on the devices of the past, no straightforward extension of those principles has been found for high heat fluxes in the smaller and more efficient electronic devices that present fabrication technologies are capable of producing. In particular, the air-cooled heatsinks of the past are ineffective in addressing the cooling needs of modern devices.

A number of alternative cooling solutions have been proposed, including liquid-based systems that pump liquid coolant across a heat dissipation apparatus. However, there are practical limitations to the efficiency of such devices when implemented at small scale. Improved cooling would allow for the use of higher-power devices and would enable an increase in the practical computing capacity of devices in many fields of technology.

SUMMARY

A method of forming heat transfer device includes forming a jet plate with a plurality of inclined jets set at an angular deviation from normal. A bottom plate is formed with channel walls that have ribs. The jet plate is attached to the bottom plate to form ribbed channels. The angular deviation of each inclined jet establishes a jet direction perpendicular to a long dimension of the ribbed channels.

These and other features and advantages will become apparent from the following detailed description of illustrative embodiments thereof, which is to be read in connection with the accompanying drawings.

BRIEF DESCRIPTION OF DRAWINGS

The disclosure will provide details in the following description of preferred embodiments with reference to the following figures wherein:

FIG. 1 is an exploded view of an embodiment of a cooling device in accordance with the present principles;

FIG. 2 is a cut-away view of an embodiment of a cooling device, providing a cross-sectional view of cooling channels, in accordance with the present principles;

FIG. 3 is a diagram of an embodiment of a coolant channel having ribs in accordance with the present principles;

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FIG. 4 is a cut-away view of an embodiment of a cooling device, providing a view along the length of the cooling channels, in accordance with the present principles;

FIG. 5 is an exploded view of an alternative embodiment of a cooling device in accordance with the present principles;

FIG. 6 is a cut-away view of an alternative embodiment of a cooling device, providing a view along the length of the cooling channels, in accordance with the present principles;

FIG. 7 is a block/flow diagram of a method for forming a cooling device according to the present principles; and

FIG. 8 is a block/flow diagram of an alternative method for forming a cooling device according to the present principles.

FIG. 9 is a diagram of an embodiment of a cooling in accordance with the present principles.

**DETAILED DESCRIPTION OF PREFERRED
EMBODIMENTS**

The present principles combine inclined impingement of coolant with ribbed cooling channels to produce a cold plate for high heat flux applications. The present embodiments use an array of impinging jets to provide high local heat transfer rates and, furthermore, provide an increased wetted surface area and increase flow turbulence along cooling channels with ribs, thereby increasing the overall heat transfer rates by exploiting efficiencies that result from particular kinds of coolant motion. When flow is turbulent, coolant particles exhibit additional transverse motion. This enhances the rate of energy and momentum exchange between coolant particles, thus increasing the heat transfer coefficient and increasing the cooling effect. This improves on conventional single-phase cooling systems, which do not concern themselves with coolant flow at such a minute level.

Although it is specifically contemplated that the present principles may be applied to fields of technology such as cooling for microprocessors and other electronic/semiconductor devices, the present principles may be applied to a broad range of devices. In particular, coolers made according to the principles described herein may be used in, e.g., industrial, automotive, and home appliance applications to more effectively dissipate waste heat. Any application that uses "active" cooling, where coolant is pumped across a heat dissipating surface, can benefit from implementing the present principles.

Referring now to the drawings in which like numerals represent the same or similar elements and initially to FIG. 1, an exemplary exploded diagram of a cold plate 100 according to the present principles is shown. A bottom channel plate 102 is shown having an inset border 103. The bottom plate 102 has a set of channel walls 104, said walls having ribs 105 along their length. Alternatively, the walls 104 may be formed on a separate plate and then joined to the bottom plate 102. The bottom channel plate 102 can be fabricated using any appropriate method or combination of methods, including milling, electrical discharge machining, etching, stamping, extrusion, and skiving. The bottom channel plate 102 should be formed from a material having a high thermal conductivity, such as copper, to promote efficient heat transfer from the device to be cooled into the coolant fluid.

A hollow side cover block 106 fits over the recessed border 103 of bottom channel plate 102. The hollow side cover block 106 forms a tight seal with the bottom channel plate 102, and may be joined using any appropriate method or combination of methods, including brazing, soldering,

and direct-bonding. The hollow side cover block **106** may be formed from the same material as the bottom channel plate **102** or may be formed from another material. Because coolant will still come into contact with the hollow side cover block **106**, the thermal transfer properties of the hollow side cover block **106** are relevant. The hollow side cover block **106** may be formed from any appropriate method, for example extrusion. The hollow side cover block **106** includes inset portions **108** to receive a jet plate **110**.

The jet plate **110** may be formed from any appropriate material. The jet plate **110** will not come into contact with the bottom channel plate **102**, but will contact channel walls **104**, such that its thermal properties will still be relevant. On the other hand, the jet plate **110** forms a border between input coolant and output coolant. A material with a lower thermal conductivity may therefore be selected, trading off heat transfer from the channel walls **104** for superior insulation between warm outlet coolant and cold input coolant.

The jet plate **110** fits into hollow side cover block **106** and joins with the channel walls **104**, forming channels with rectangular cross sections. Alternative cross sections may be formed based on the disposition of the channel walls **104**. It is specifically contemplated that quadrilateral cross-sections will be used, but other shapes may also be employed, such as triangles. The bottom surface of jet plate **110** has rows of inclined jet ports **112**, which open into the channels formed by the jet plate **110** and the bottom channel plate **102**.

A top cover plate **114** fits onto the hollow side cover block **106** and the jet plate **110**, forming an inlet plenum. The top plate **114** may be formed of any appropriate material. In particular, the top plate **114** may be formed from materials similar to those employed in other components, or may be formed from a material having less thermal conductivity. A material with lower thermal conductivity is advantageous in the top plate **114**, as such a material provides superior insulation between warm outlet coolant and cold input coolant, as well as inhibiting the infiltration of ambient heat from the environment. The top cover plate has outlet ports **116** and an inlet port **118**. The number and particular positioning of outlet and inlet ports **116** and **118** may be altered according to the needs of the particular application.

Coolant may be introduced and removed from cold plate **100** using any appropriate form of pump. The coolant may then be run through a larger heat-dissipation unit, e.g., a heat sink (not shown), reducing the temperature. This allows coolant to be recycled through the cold plate **100** in a closed system.

Referring now to FIG. 2, a cut-away view of an assembled cold plate **100** is shown, illustrating coolant circulation. The coolant may be any suitable material, including air, water, and oil. Exemplary coolants include deionized water, propylene glycol, and mineral oil. The coolant material should be chosen keeping in mind the constituent materials of cold plate **100**, to prevent chemical reactions or degradation from occurring. An advantageous coolant will have a high thermal conductivity, high thermal capacity, and low viscosity. Coolant, illustrated by arrows, enters plenum **202** through inlet hole **118**. The inclined jet holes **112** are shown in jet plate **110**, guiding coolant into channels **204** at an angle, causing coolant to circulate within the channels **204**. This angle is directed perpendicular to the flow of coolant through the channels **204**, producing circulatory turbulence.

The entrance effect in the inclined jets **112** causes local acceleration of the coolant jets and increases thereby the local heat transfer rates as compared to a normally impinging jet **112** in a combined environment. This effect is partially due to an increased coolant velocity transverse to

the channel direction, such that any given quantity of coolant will come into contact with more of the internal surface area of channel **204**. The vortex that results in the coolant is significantly more effective in transferring heat away from the surface of channels **204** than would result with a normally impinging jet **112**, and it is clear that additional turbulence increases the heat transfer rate further.

The number of jets **112**, jet impingement angle, jet diameter, jet location, and jet orientation may be determined according to the needs of the particular application. Some exemplary impingement angles include 20, 30, and 45 degrees with respect to the normal. It should be recognized that any impingement angle may be used, although too large of an angle will impede coolant flow. An exemplary diameter for jets **112** could be about 1 mm, and the jet **112** could be oriented to impinge upon either of the channels' **204** sidewalls or upon the channel's base.

It should also be recognized that the jets **112** may have any appropriate cross-sectional shape. Although it is specifically contemplated that the jets **112** may have a circular cross-section, other shapes may also be employed, including square and triangular cross-sections. One advantageous spacing for jets **112** is three times the diameter of the jets **112**, although it is contemplated that other spacings may be employed within the present principles. Placement, orientation, and size of jets **112** should be selected to maximize coolant flow and turbulence, such that heat transfer in the channels **204** is maximized.

It should be noted that channels **204** have ribs **105** on three of its four walls. Ribs **105** may be used on one, two, three, or all of the walls of channels **204**, depending on the intended application, but it should be noted that having ribs on the top wall of channels **204**, e.g., on the jet plate **110**, will not contribute substantially to heat transfer and will increase the pressure drop across the cold plate **100**. As such, having ribs **105** on the jet plate **110** may adversely affect the overall dissipative performance of the cold plate **100**. Ribs **105** keep the flow in the channels **204** turbulent and increase wetted surface area. As those having skill in the art will recognize, wetted surface area refers to the surface area of the solid bulk of the walls of channels **204** that is in contact with the coolant fluid. This increase of surface area increases the rate of heat transfer even if a "dry" coolant, such as air, is used.

Ribs **105** may be formed by milling, electrical discharge machining, etching, stamping, machining, extrusion, or any other suitable technique. Although for simplicity it is specifically contemplated that the ribs will be formed from the same material as the walls of the channels **204**, the ribs **105** may be formed from any appropriate material and added to the channels **204** in any appropriate fashion. Furthermore, ribs may be formed in any shape. For example, the ribs may have a triangular, rectangular, or cylindrical cross section. Furthermore, rib spacing may be adapted to the needs of the particular application and coolant and may be consistent or may vary from rib to rib.

The bottom channel plate **102** is placed into contact with a surface of a heat source (not shown). Heat is carried away from the heat source and into bottom channel plate **102** by conduction. Because imperfections in solid surfaces may decrease thermal contact and inhibit heat flow, heat transfer may be facilitated by using a substance to promote thermal contact, such as thermal paste, which conducts heat well and which makes contact between the heat source and bottom channel plate **102**.

Referring now to FIG. 3 is a cutaway view of a channel **204**. As can be seen, there are ribs **105** along the bottom and

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two sides of the channel **204**, but not along the top of the channel **204**. The ribs **105** are shown as being regularly spaced, though it is contemplated that the spacing between the ribs **105** may vary. One exemplary profile for the ribs **105** could have a rib height of about 10% of the channel width and a rib spacing of about 5-10 times the rib height. In addition, ribs on multiple walls of the channels **204** could be arranged in an inline or in a staggered manner.

Ribs **105** may be formed with any appropriate cross-sectional shape. In particular, rectangular cross-sections are shown in FIG. 3, but it is contemplated that other shapes, including triangles and semicircles, may be used. The ribs **105** increase coolant turbulence, and “rougher” surfaces will cause greater increases in turbulence. As such, cross-sections with sharp edges or irregularities are better suited to being used in ribs **105**. It is also contemplated that ribs **105** may be formed in such a way as to produce random imperfections on the rib surface, such as machining marks, that further increase coolant flow turbulence.

Referring now to FIG. 4, a cutaway view of an assembled cold plate **100** is shown, viewing the length of one of the channels **204**. Coolant enters by way of coolant input **118** into plenum **202**. Coolant then passes through inclined impingement jets **112** into the channel **204**, creating turbulent flow. The ribs **105** are not shown in this figure for the sake of simplicity. The coolant then passes through side cavities **402** and exits by the coolant outputs **116**.

FIG. 4 illustrates the fit between the bottom channel plate **102** and the hollow side cover block **106**.

Referring now to FIG. 5, an alternative embodiment of a cold plate **500** is shown. A channel plate **502** includes channel walls **504** with ribs. The channel walls **504** may be milled, electrical discharge machined, etched, stamped, skived, machined, or formed through some other appropriate process. Alternatively, the channel walls **504** may be separately formed and attached to channel plate **502**. Channel plate **502** includes a coolant output port **506**. Although the coolant output port **506** is depicted as being disposed within a side wall of the channel plate **506**, the coolant output port **506** may alternatively be disposed in a bottom surface of the channel plate **506**.

A jet plate **508** is attached to the top of channel plate **502** and has a set of jets **510**. As above, the jets **510** may be inclined to increase coolant turbulence and heat transfer. A cover plate **512** is placed on top of the jet plate **508** and has a coolant intake port **514**. The parts can be joined to one another by any appropriate process, including brazing, soldering, and direct-bonding.

Referring now to FIG. 6, a cut-away view of cold plate **500** is shown, illustrating coolant flow. Coolant enters through intake port **514** into a plenum **602**. Plenum **602** circulates coolant to the jets **510**, which provide the coolant to channels **604**. The channels **604** may extend along the entire length of the plenum **602** or may be shorter. In the present embodiment, the channels **604** are shorter than the full length of plenum **602**, so that there are no jets **510** directly beneath the coolant intake **514**. This provides greater distribution of coolant across the jets **510** along the channel. This improves the uniformity of cooling. If the coolant enters predominately at certain jets **510**, the cold plate **500** directly beneath those jets **510** will receive a disproportionate amount of cooling. By placing the intake **514** over a section that does not have jets **510**, the coolant is forced to circulate more uniformly through the plenum **602**.

Referring now to FIG. 7, a block/flow diagram is shown for forming a cooling device according to the present

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principles. Block **702** forms a top cover plate **114** that has at least one inlet port **118** and at least one outlet port **116**. The top cover plate **114** may be formed by any appropriate process, including for example milling, electrical discharge machining, etching, stamping, and extrusion. Block **704** forms the jet plate **110** including jets **112**. The jets **112** may be formed with the jet plate by the above processes, or they may be drilled into the jet plate **110**. Block **706** forms side cover block **106**, having guides **108** of such a size as to accommodate the jet plate **110**. Block **708** forms the cooling plate **102** having channel walls **104**. The channel walls **104** may be formed integrally with the cooling plate **102** by, e.g., skiving, or may be formed on a separate surface and then attached to cooling plate **102** by any appropriate method or combination of methods, including brazing, soldering, and direct-bonding.

Block **710** fits the side cover block **106** onto the cooling plate **102**. The cooling plate **102** may be designed with a recessed border **103** to accommodate the side cover block **106**, such that the side cover block **106** and the cooling plate **102** form a tight seal to prevent coolant leakage. Block **712** fits the jet plate **110** into the guides **108** in the side cover block **106**, and block **714** first the top plate onto the side cover plate **106** and jet plate **110**, forming plenum **202**. The points of contact between the various pieces are joined in such a way as to prevent coolant leakage along the edges, which would diminish cooling efficiency.

Referring now to FIG. 8, a block/flow diagram is shown for an alternative method of forming a cooling device according to the present principles. Block **802** forms a top cover plate **512** that has at least one inlet port **514**. Again, the elements of this embodiment may be formed by any appropriate process, including for example milling, electrical discharge machining, etching, stamping, and extrusion. Block **804** forms jet plate **508** with jets **510** being formed integrally with the jet plate **508** or added afterward by machining or drilling. Block **806** forms a cooling plate **502** with channel walls **504** and sidewalls. As above, the channel walls **504** may be formed integrally with the cooling plate **502** or may be formed separately and then attached to the cooling plate **502** by, e.g., brazing, soldering, and direct-bonding.

Block **808** forms one or more outlet ports **506** in the sidewalls of cooling plate **502**. The outlet port **506** may optionally also be formed integrally with the cooling plate **502** in block **806**. Block **810** fits the jet plate **508** onto the cooling plate **502**, and block **812** fits the top plate **512** onto the jet plate. The points of contact between the various pieces are joined in such a way as to prevent coolant leakage along the edges, which would diminish cooling efficiency.

Referring now to FIG. 9, a diagram of an exemplary cooling system is shown. Chips **904** are shown, which generate heat during operation that must be removed. Each chip **904** has a cold plate **902** attached to it. The cold plate **902** may be attached using a thermal interface material to promote heat transfer away from the chip **904**. Coolant fluid is then pumped through the cold plate **902** and circulated to a heat rejection unit **906** which removes heat from the coolant and releases it into, e.g., the environment. The coolant then returns to the cold plates **902** to be used again.

Having described preferred embodiments of a cold plate with combined inclined impingement and ribbed channels (which are intended to be illustrative and not limiting), it is noted that modifications and variations can be made by persons skilled in the art in light of the above teachings. It is therefore to be understood that changes may be made in the particular embodiments disclosed which are within the

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scope of the invention as outlined by the appended claims. Having thus described aspects of the invention, with the details and particularity required by the patent laws, what is claimed and desired protected by Letters Patent is set forth in the appended claims.

What is claimed is:

1. A method for forming a heat transfer device, comprising:

forming a jet plate with a plurality of inclined jets set at an angular deviation from normal;
forming a bottom plate with channel walls that have ribs;
and

attaching the jet plate to the bottom plate to form a plurality of ribbed channels, wherein the angular deviation of each inclined jet establishes a jet direction perpendicular to a long dimension of the plurality of ribbed channels.

2. The method of claim 1, further comprising forming one or more outlet ports in a sidewall of the bottom plate.

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3. The method of claim 1, further comprising:
forming a side cover block; and
attaching the jet plate to the side cover block.

4. The method of claim 3, wherein attaching the jet plate to the side cover block comprises a joining process selected from the group consisting of brazing, soldering, and direct-bonding.

5. The method of claim 1, wherein each ribbed channel has ribs on only three of four channel walls.

6. The method of claim 1, wherein the jet plate is formed with a material with a lower thermal conductivity than a material of the bottom plate.

7. The method of claim 1, wherein plurality of inclined jets comprises a respective plurality of inclined jets for each of the plurality of ribbed channels.

8. The method of claim 7, wherein each respective plurality of inclined jets has a spacing between inclined jets of about three times a diameter of the inclined jets.

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